Graduate Research

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Disinfection byproducts (DBPs) are formed when disinfectants like chlorine react with natural organic matter, bromide, or synthetic pollutants in drinking water during water disinfection process. Current studies have found that DBPs regulated by regulations only account for a small portion of the total DBP abundance. A large number of unknown and unregulated DBPs have a stronger toxic potential[1].

Emerging DBPs, including nitrogenous compounds and halogenated quinones, are reported to induce strong genotoxicity, oxidative stress, and cell apoptosis at nanomolar levels[2]. DBP exposure has also been associated with adverse health effects like cancer, reproductive issues, and developmental disorders[3]. While their impacts on general human health have been studied, their specific effects and underlying mechanisms on the human gut microbiota remain largely unknown. Therefore, it is essential to understand how DBPs being ingested from drinking water would affect human gut microbiome to evaluate long-term health risks and provide implications to drinking water treatment technology.

1. Research preparations

Since Autumn Quarter 2024, I have been involved in a research project investigating whether drinking water disinfection by-products (DBPs) would affect human gut microbiome composition, potentially leading to inflammation and adverse health outcomes. Beginning in Winter Quarter, I have actively participated in experiments in both Professor Justin's lab and Professor Mitch's lab.

In January 2025, I engaged in technical training across two research laboratories to strengthen my experimental skills in both chemical analysis and microbial biotechniques. In the Mitch Lab, I focused on learning and practicing water chloramination disinfection (widely used in drinking water disinfection treatment) and disinfection byproduct (DBP) quantification, where I learned to conduct liquid-liquid extraction using varying concentrations of standard DBP solutions. The extracted samples were prepared in GC vials for subsequent gas chromatography analysis, providing hands-on experience with water quality assessment protocols. At the same time, in the Justin Lab, I received training in anaerobic microbial culturing, including the operation of an anaerobic incubator and the inoculation of strict anaerobes into a 24-well plate system. This experience deepened my understanding of microbial cultivation under anaerobic conditions and prepared me for future microbiological research by strengthening my grasp of the experimental topic.

During February and March 2025, I continued cross-disciplinary training in both the Justin and Mitch labs. In the Justin Lab, I continuously contributed for four weeks to the construction of the hCom2[4] synthetic microbiome by extracting and purification of DNA from various microbial colonies for downstream sequencing verification.

In Mitch's lab, I practiced chlorination using both upstream and downstream wastewater treatment samples and received hands-on training in operating the GC-MS system for chemical composition and DBP concentration analysis. In addition, I actively participated in a technical walkthrough of the water disinfection workflow and DBP quantification procedures in collaboration with Yifan, with support from Mitch lab members Vincent and Ben.

2. Research Method

The final formal experimental water location was decided in Boronda Lake, and the time for experiment would begin at May 2025 and end at July 2025. The experiment time line is shown in Figure 1.

As part of the project outcomes, I prepared the experimental protocol as well as simplified laboratory procedures for water treatment and DBP measurement, which were later revised and modified by Yifan (Figure 2).

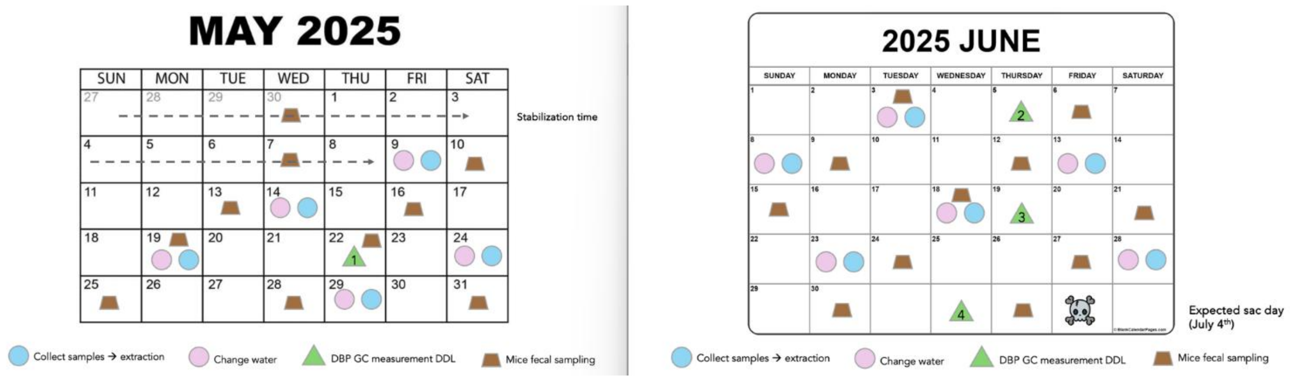


Figure 1. The experiment timeline for May and June 2025.

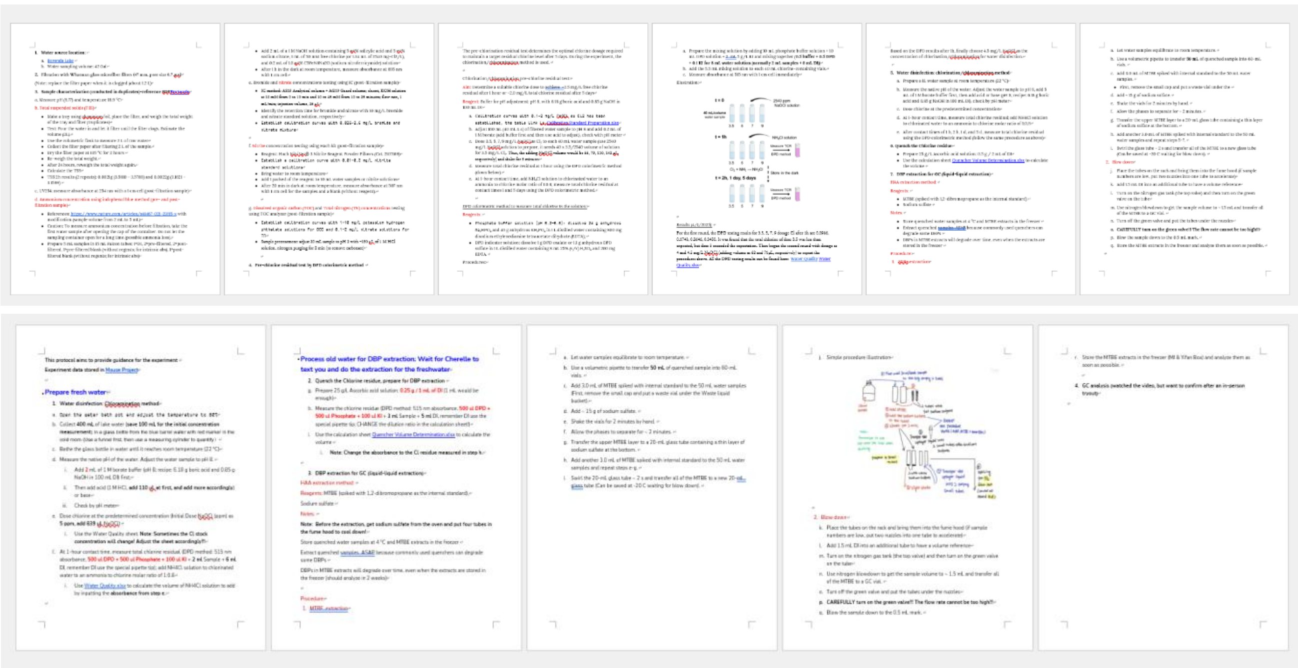


Figure 2. The experimental protocols.

3. Preliminary Research Results and Discussions

From March 2025 to the present, as the formal experiment began, I have continued working with Yifan on water preparation and replacement, as well as the analysis of DBP concentrations in disinfection water samples.

3.1. Composition of DBPs from water chloramination and specific DBP accumulation over time

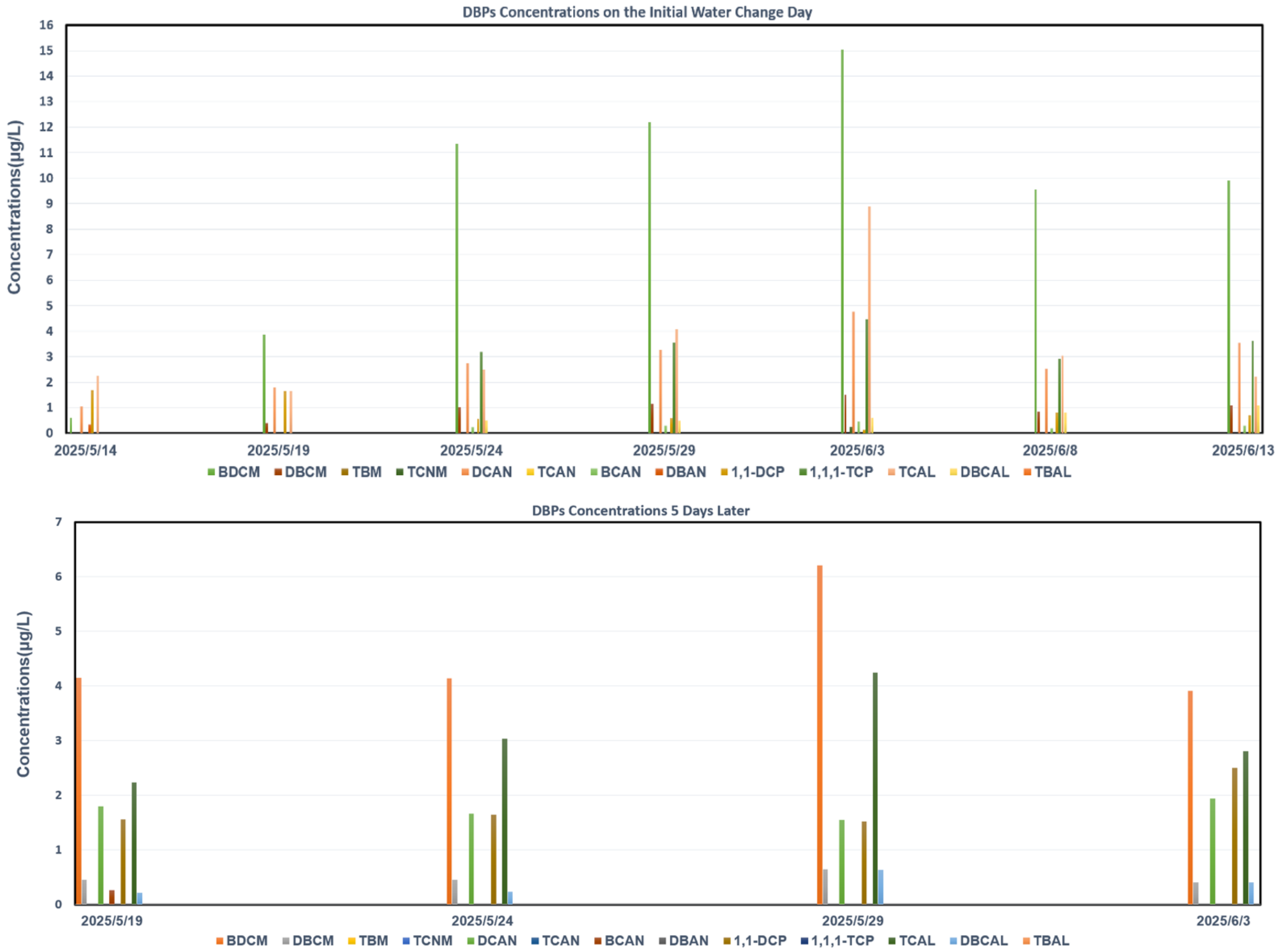
The DBP concentration was measured at the time of the initial water change and again 5 days later, during the next water change, to track its variation. Information of DBPs tested in the experiment were showed in Table 1.

The concentrations of DBPs measured on the initial water change days and again five days later, at the next water change, are shown in Figure 3. The results indicate that TCM, BDCM, 1,1,1-TCP, TCAL, DCAN were the dominant DBPs formed during water chloramination disinfection, with TCM present at significantly higher levels. After 5 days, most DBPs showed a marked decline, with compounds such as BCAN and 1,1,1-TCP decreasing to undetectable levels, indicating overall DBP loss. In contrast, 1,1-DCP exhibited an opposite trend, with its concentration increasing over time, which may result from the transformation of intermediate compounds or slower hydrolysis kinetics under the storage conditions. These results may help identify the primary contributors to DBP-associated toxicity.

DBP Class		DBP	Abbreviation
		1-2 Carbon DBPs (or Alliphatic DBPs)	
Trihalomethanes	THMs	Trichloromethane	TCM
		Bromodichloromethane	BDCM
		Dibromochloromethane	DBCM
		Tribromomethane	TBM
Haloketones	HKs	1,1-dichloropropanone	1,1-DCP
		1,1,1-trichloropropanone	1,1,1-TCP
Haloacetaldehydes	HALs	Trichloroacetaldehyde	TCAL
		Dibromochloroacetaldehyde	DBCAL
		Tribromoacetaldehyde	TBAL
Haloacetonitriles	HANs	Dichloroacetonitrile	DCAN
		Trichloroacetonitrile	TCAN
		Dibromoacetonitrile	DBAN
		Tribromoacetonitrile	TBAN
Halonitromethanes	HNMs	Trichloronitromethane (commonly referred to as chloropicrin)	TCNM

Table 1. The detailed information of DBPs measured in the experiment.

(A)



(B)

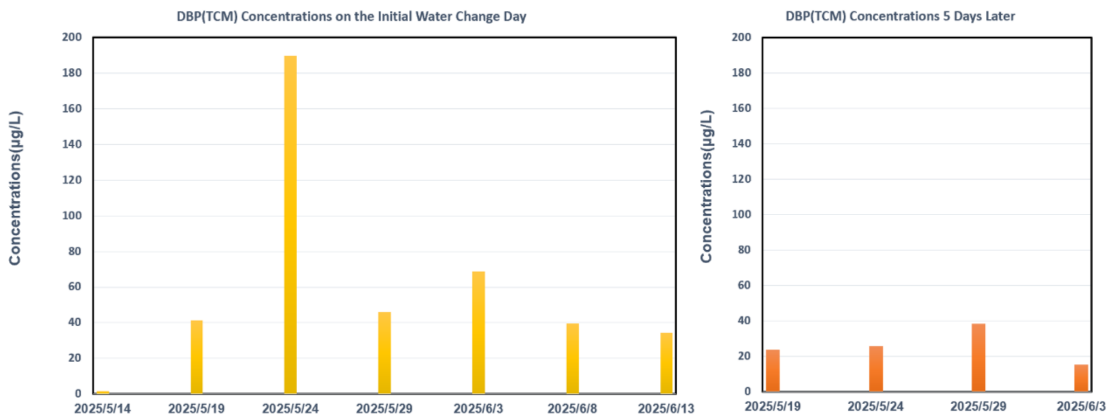


Figure 3. A and B. Concentrations of various DBPs (except TCM) in water disinfection samples on the initial day and after 5 days (A). Given its relatively high concentration, TCM was analyzed separately on the initial day and again five days later (B). Standard errors were not shown in the bar chart due to limited replicate

numbers in each group.

3.2. DBPs cause potential negative effects on synthetic culturing gut microbiome

The microbial community testing yielded results, as shown in Figure Z. In this figure, the gut microbial community derived from an Asian stool sample was tested, including exposure to four different water treatment groups (lake and MilliQ water with DBP addition, original lake and MilliQ water as control). After treated with DBP, the composition of Asian gut microbiome had changed. On the one hand, the *Prevotellaceae* group was initially abundant, but its abundance (shown in red) significantly declined in the DBP-treated lake water group, indicating that DBPs had an inhibitory effect on its growth. On the other hand, the *Bacterodaceae* group showed an increase in abundance of DBP treated group. The low abundance of *Prevotellaceae* and the high abundance of *Bacterodaceae* were distinctive features of western gut profiles, suggesting that disinfection by-products, particular for nature lake water may promote a shift in the Asian community toward a westernized composition.

The hunter-gatherer community from Hadza tribe in Africa and western guts were also tested. The results indicated that the Hadza *Prevotellaceae* (*Prevotella copri*) growth was greatly inhibited along with the increase of DBP concentrations, while the same species from western gut didn't, further supporting the assumptions.

In summary, the first-round testing indicated that the drinking water disinfection by-products (DBPs) would negatively affect the human gut microbiome.

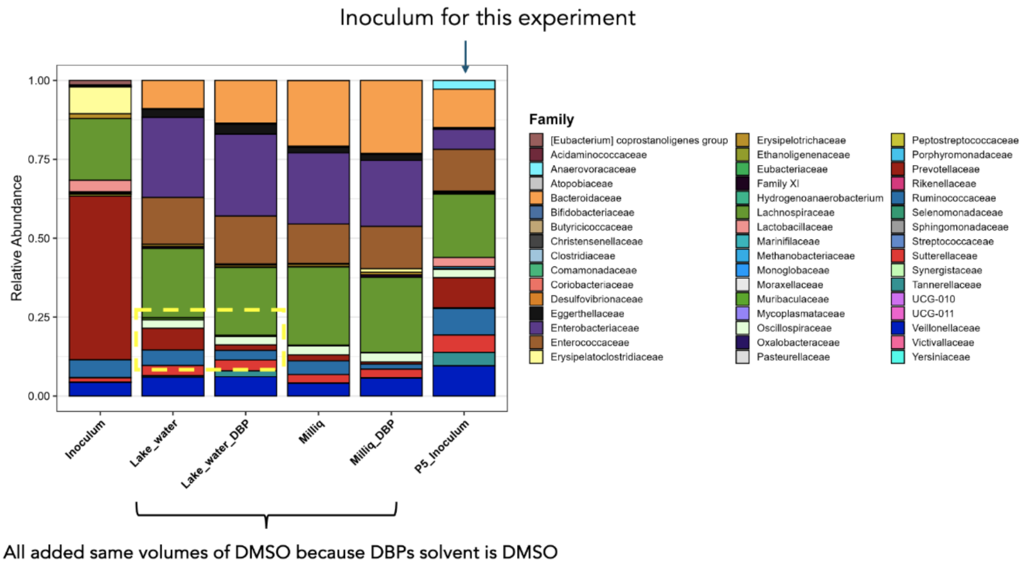


Figure 3. The culturing microbial community composition treated with different water samples.

4. Future work

The mouse experiment is still ongoing and is expected to conclude by the end of June. Fecal samples will be sequenced, and tissues including blood, colon, and small intestine will be examined. Once the mice data are obtained, they will be analyzed align to the DBP concentration data to explore potential associations between DBPs exposure and alterations in the gut microbiome. Additionally, possible underlying mechanisms will be evaluated. These analyses will generate additional data to further support and explore our assumptions and the project.

References:

1. Lau, S. S. *et al.* Cytotoxicity Comparison between Drinking Water Treated by Chlorination with Postchloramination versus Granular Activated Carbon (GAC) with Postchlorination. *Environ. Sci. Technol.* **57**, 13699–13709 (2023).

2. Lau, S. S. *et al.* Toxicological assessment of potable reuse and conventional drinking waters. *Nat. Sustain.* **6**, 39–46 (2023).

3. Li, X. *et al.* A Review of Traditional and Emerging Residual Chlorine Quenchers on Disinfection By-Products: Impact and Mechanisms. *Toxics* **11**, 410 (2023).

4. Cheng, A. G. *et al.* Design, construction, and *in vivo* augmentation of a complex gut microbiome. *Cell* **185**, 3617-3636.e19 (2022).